

control/ decision/ conditional statments

Single line approach

Var = stmt_1 if condtion_1 else Stmt2 if condtion2
else stmt_3 if condtion3 else stmt4

```
In [1]: # a=int(input("enter the a value:-"))
b=int(input("enter the b value:-"))
res= "Equal" if a==b else "NOt Equal"
print(res)
```

NOt Equal

```
In [8]: a= int(input( "enter the a value:-"))
b= int(input( "enter the b value:-"))
var= "equal" if a==b else "a is big" if a>b else "b is big"
print(var)
```

b is big

```
In [14]: a= int( input( 'enter the a value:- '))
b= int( input( 'enter the b value:- '))
c= int( input(" enter the c value : -"))
vars= "equal" if a==b==c else "a is big " if a>b else " b is big " if b>c else "
print(vars)
```

a is big

Using AND and OR in conditional statments

```
In [17]: a=int(input( 'enter a value:-'))
b=int(input( 'enter b value:-'))
res= (a==b) and "equal" or "not equal"
print(res)
```

not equal

```
In [26]: a=int(input( 'enter a value:-'))
b=int(input( 'enter b value:-'))
c=int(input( 'enter c value:-'))

res= (a==b) and "equal" or ( a>b) and " a is big " or (b>c) and "b is big " or "
print(res)
```

c is big

multi line approch

simple if

```
In [4]: a= int (input("enter a value:-"))
        b= int (input("enter b value:-"))

        if a==b:
            print("equal")
            print("thank you")

        print("bye")
```

bye

```
In [6]: a= int(input("Enter a value:- "))
        b= int(input("Enter b value :-"))
        if a==b:
            print("equal")
            print("Thank you")
        if a!=b:
            print("Not equal")
            print("bye")
```

Not equal

bye

if -else

```
In [8]: n=int(input("enter the number"))
        if n%2==0:
            print("even")
        else:
            print("odd")
```

odd

```
In [12]: import calendar as c
        year= int(input("enter year :-"))
        if c.isleap(year):
            print(year,"is a leap year")
        else:
            print(year,"is not a leap year")
```

2001 is not a leap year

```
In [15]: c.isleap(2000)
```

Out[15]: True

```
In [17]: import math
        n=int(input("enter a value"))
        m=math.sqrt(n)
        if m*m==n:
            print("it is a perfect sqrt root ")
        else:
            print("not a perfect sqrt")
```

not a perfect sqrt

if elif esle

```
In [2]: a= int(input("enter a value"))
        b= int(input("enter b value"))

        if a==b:
            print("BOTH ARE EQUAL")
        elif a>b:
            print("enter the a is bigger")
        else:
            print("enter b is big")
```

enter b is big

```
In [24]: a= int(input("enter a value"))
        b= int(input("enter b value"))
        c= int(input("enter c value"))

        if a==b==c:
            print("all ARE EQUAL")
        elif a>b and a>c and a!=b:
            print(" a is bigger")
        elif b>c:
            print(" b is big")
        else:
            print("c is big")
```

b is big

Nested if

```
In [26]: n=int(input('enter n value:'))
        if n>0:
            print("it is a +postive")
            if n%2==0:
                print("even")
            else:
                print("odd")
        else:
            print("it is -ve")
```

it is a +postive
even

In []: